

AN **INDIAN** ADAPTATION

Third Edition

Supply Chain Management

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WILEY



SUBJECT: SUPPLY CHAIN MANAGEMENT STRATEGY

CHAPTER: 08

CHAPTER TITLE: Forecasting and Demand Planning

REFERENCE BOOK:

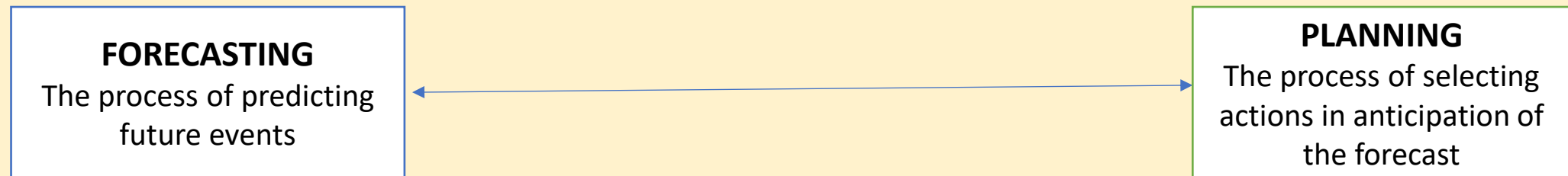
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Lecture 23

Forecasting Vs. Planning

- **Forecasting** is the process of predicting future events.
 - It is a systematic prediction of future demand to help firms plan production, inventory, workforce, and distribution efficiently.
- **Planning** is the process of selecting actions in anticipation of the forecast.



Forecasting

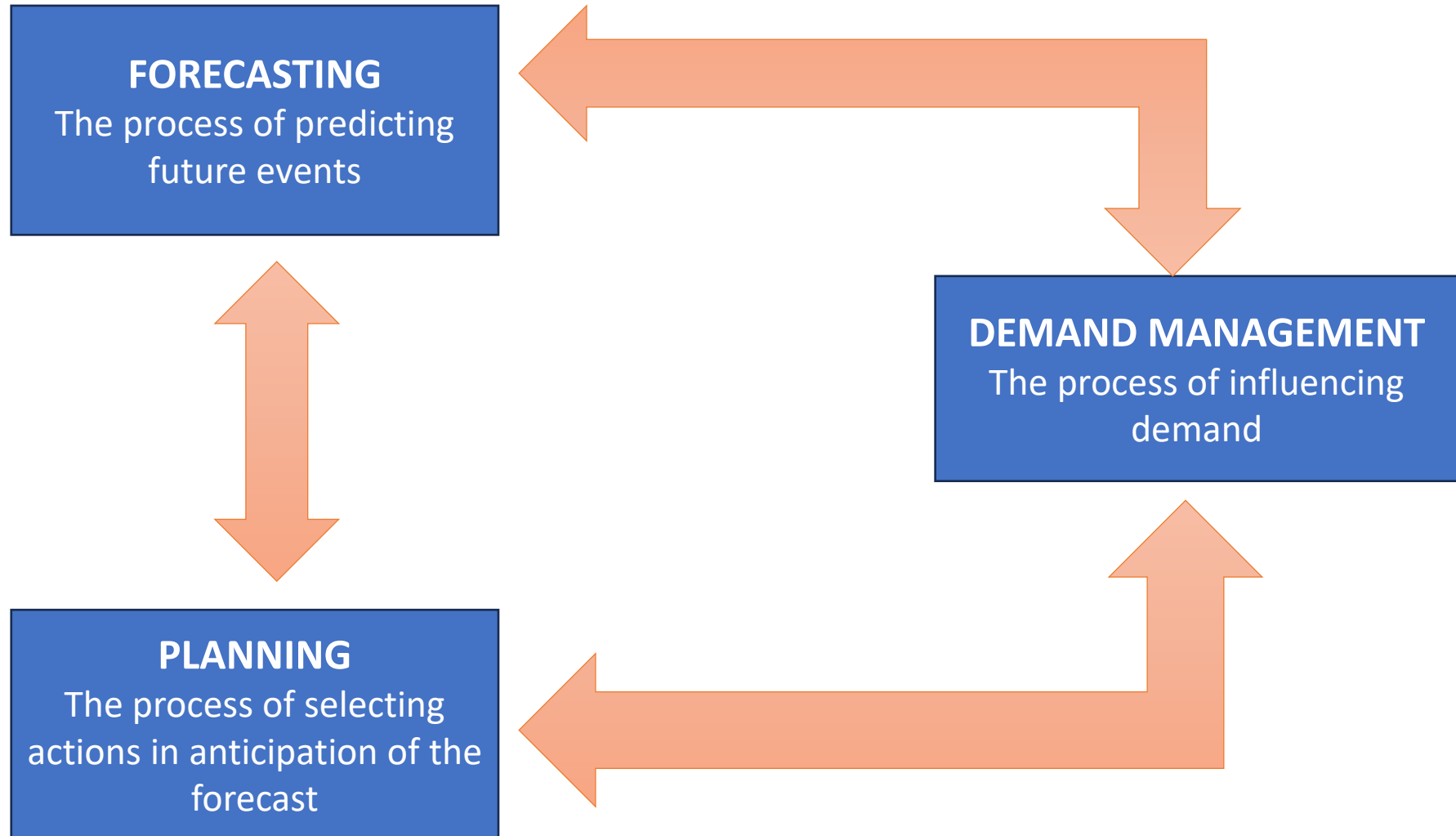
- Forecasting is one of the most important business activities, as it drives all other business decisions.
- Decisions such as which markets to pursue, which products to produce, how much inventory to carry, and how many people to hire are based on a forecast.
- Forecasting is a challenge.
- Poor forecasts result in poor planning and can leave the company unprepared to meet future demands.
- The consequences can be costly in terms of lost sales or excess inventory that can not be sold.

Planning

Planning involves:

- 1) Scheduling existing resources:** For a business and its supply chain to be competitive, it must use its current resources in the most efficient way possible. This includes the production process, transportation, labour, facilities, and capital.
- 2) Determining future resource needs:** In addition to efficiently utilising current resources, organisations must determine what resources are going to be needed in the future. These decisions depend on forecasts of emerging market opportunities, new technology, new products, and competition.
- 3) Acquiring new resources:** It takes time for companies to acquire new facilities, new technologies, and new equipment or expand to new locations. Plans must be made well in advance, and procedures to acquire new resources and capabilities put in place well ahead of time.

Forecasting, Planning and Demand Management



Forecasting, Planning and Demand Management

- Companies confuse forecasting and planning.
- The reason is that companies have the ability to affect demand.
- This can be done through promotional campaigns and advertisements, offering incentives to sales staff and personnel, and cutting prices. This is called **Demand Management** and is the process of influencing demand.

Impact of Forecasting on Supply Chain Management (SCM)

- The forecast of demand affects all the plans made by each company in the chain.
- When members of the supply chain make their forecasts independent of one another, they are only looking at the demand of their immediate buyer not the final customer in the chain.
- The consequences are a mismatch between supply and demand because each member of the chain is working to fulfill a different level of demand.

Forecasting Process

- Forecasting process involves certain principles. These are:
 - 1) **Forecasts are rarely perfect**
 - 2) **Forecasts are more accurate for groups than for individual items**
 - 3) **Forecasts are more accurate for shorter than longer time horizons**

Principles of Forecasting

1) Forecasts are rarely perfect:

- A perfect forecast is rare as there are too many factors in the business environment that cannot be predicted with certainty.
 - Forecasting the future involves uncertainty and forecasters know that they have to live with a certain amount of error, which is the difference between the forecast and what actually happened.

Principles of Forecasting

2) Forecasts are more accurate for groups than for individual items:

- When items are grouped together, such as in product groups or families of items, their individual high and low values cancel each other out.
- The data for a group of items can be stable even when individual items in the group are unstable.
- The result is that higher degrees of accuracy can be obtained when forecasting for a group rather than for individual items.

Principles of Forecasting

3) Forecasts are more accurate for shorter than longer time horizons:

- The shorter the time horizon of the forecast, the lower the degree of uncertainty.
- Data do not change much in the short run.
- As the time horizon increases, however, there is a much greater likelihood that changes in established patterns and relationships will occur.
- For that reason, long-range forecasting tends to be less accurate than short-range forecasting.

Steps in Forecasting Process

1) Decide what to forecast:

- Forecasts are made to help plan for the future. First, to decide what forecasts are actually needed to guide the plan

2) Analyse appropriate data:

- This step involves analysing available data and identifying patterns present.
- Most common data patterns are
 - a) Level or horizontal: When data fluctuate around a constant mean. It is the easiest to predict and is common for commodity products in the mature stage of the life cycle, such as table salt or toothpaste.
 - b) Trend: When data exhibit an increasing or decreasing pattern over time (the simplest type of trend is a straight line or linear line)
 - c) Seasonality: Any pattern that regularly repeats itself, such as restaurant sales that peak on the weekend.
 - d) Cycles: These patterns are created by economic fluctuations. Examples include recessions, inflation, or even the life cycle of a product.

Steps in Forecasting Process

- In addition to these data, the data contain random variation that cannot be predicted.
- The more random variation in the data, the harder it is to forecast.

3) Select the forecasting model:

- After identifying the data patterns, the next step is to select an appropriate forecasting model.

4) Generate the forecast:

- Once a model has been selected, the forecast is generated.

5) Monitor forecast accuracy:

- Forecast accuracy should be regularly monitored by measuring forecast errors so that the forecasting process can be continuously improved as new information and data become available.

Factors in Method Selection

1. Amount and type of available data:

- The accuracy and choice of a forecasting method depend on how much historical data you have and whether it is qualitative (expert opinions, market insights) or quantitative (numerical sales, demand, or time-series data).

2. Degree of accuracy required:

- The level of precision needed in the forecast determines which forecasting method to use, since some decisions require highly accurate predictions while others can work with approximate estimates.

3. Length of forecast horizon:

- The time span into the future for which predictions are made, which influences the forecasting method and expected level of accuracy.

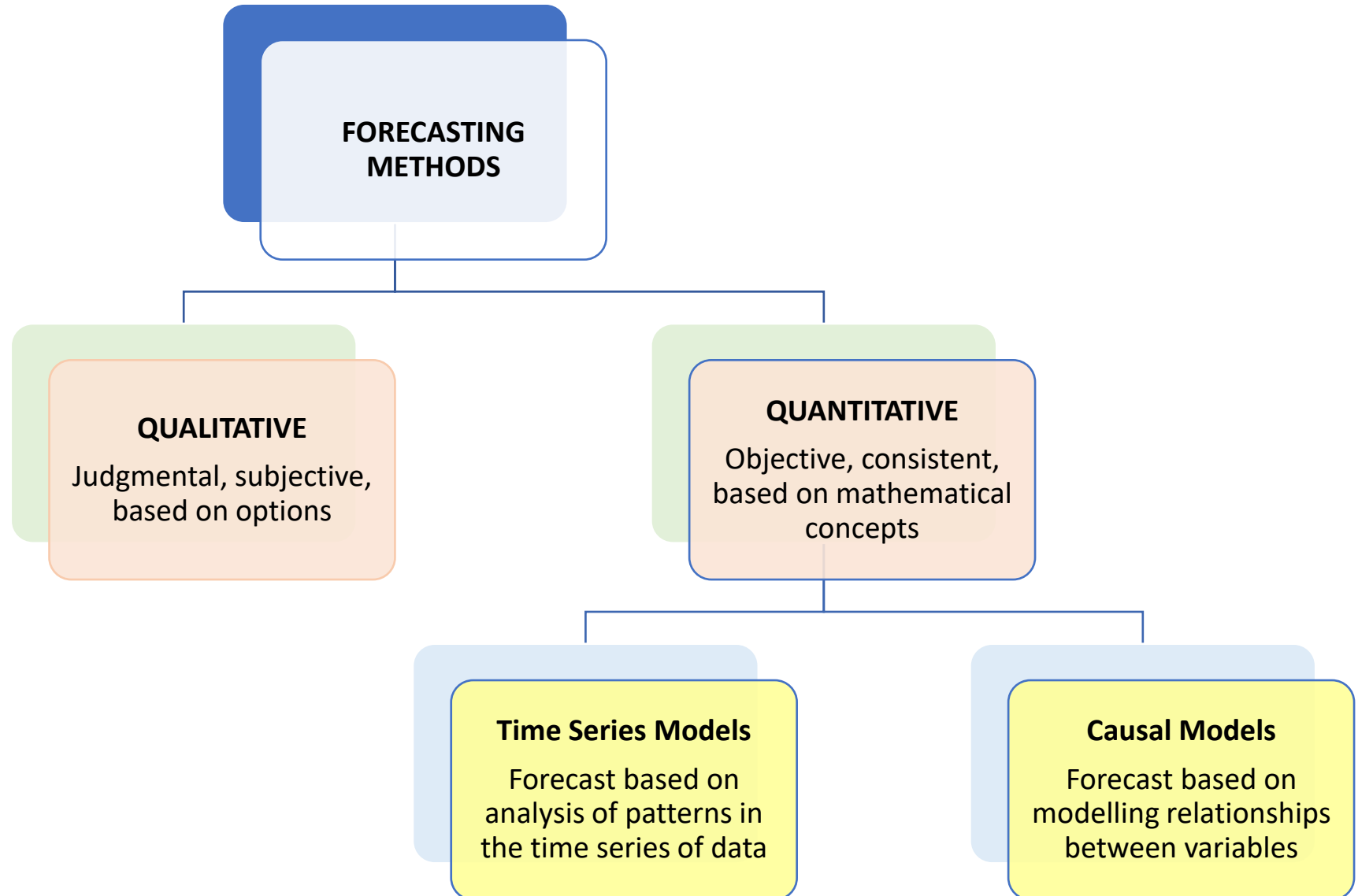
4. Patterns in the data:

- Identifying trends, seasonality, cycles, or random variations in past data to choose the most suitable forecasting method.

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Lecture 24

Types of Forecasting Methods



Forecasting Methods

Qualitative Forecasting Methods:

- Based on subjective opinions and judgments of individuals, such as managers, sales staff, or customers.
- One type of qualitative method is “**intention surveys**”, which means asking customers whether they would buy a particular product.
- Another type of method is “**sales force composite**”, which means when the sales staff make a group forecast of upcoming sales.
- These methods are made by people and, as a result, are subjective to human biases.

Forecasting Methods

Quantitative Forecasting Methods:

- Based on mathematical modeling.
- These methods are objective and consistent, are capable of handling large amounts of data, and can uncover complex relationships.
- Provided that good data are available, these methods are generally more accurate than qualitative methods.

Time Series Forecasting Models

- A time series is simply a listing of data points of the variable being forecast over time taken at regular intervals.
- For example: Census data over the past 50 years.

(1) Mean:

- The simplest forecasting model is the mean.

$$F_{t+1} = \frac{\sum D_t}{n}$$

or

$$F_{t+1} = \frac{[D_t + D_{t-1} + \dots + D_{t-n}]}{n}$$

Where, F_{t+1} = Forecast of demand for next period, t+1

D_t = Demand for current period, t

n = Number of data points

- A reasonable forecasting model to use for forecasting stable and mature products.

Time Series Forecasting Models

Question:

Handy power tool is forecasting sales of its popular drill. The company has been selling the drill for the past 15 years, and sales have been steady. The company uses a simple mean to forecast weekly sales. Sales over the past five weeks are available, and the company will use them to make a forecast for week six.

Answer:

The forecast is simply an average of the past five weeks:

$$F_{t+1} = \frac{[D_t + D_{t-1} + \dots + D_{t-n}]}{n}$$

$$\text{Or } F_6 = \left[\frac{8+10+9+12+10}{5} \right] = \frac{49}{5} = 9.8 = 10$$

Week	Drill Sales
1	8
2	10
3	9
4	12
5	10

Problem:

Suppose the demand for a product in the last 4 months (January- April) is given, then find out the demand for May?

Month	Demand
January	100
February	120
March	110
April	130

Time Series Forecasting Models

(2) Moving Average:

- It generates a forecast by averaging a specifies number, n , of the most recent data rather than the entire data set.
- As new data become available, the oldest are dropped, and the number of observations used to compute the average is kept constant.
- The simple moving average “moves” through time.

$$F_{t+1} = \frac{\sum D_t}{n}$$

or

$$F_{t+1} = \frac{[D_t + D_{t-1} + \dots + D_{t-n}]}{n}$$

Where, n = Number of data points in the moving average

Time Series Forecasting Models

Question:

Ramesh forecasts sales of kitchen aprons using a three-period moving average. How many aprons should they expect to sell in April given the following sales figures from January to March? To find the forecast for April we take an average of the last three observations

Answer:

$$F_{t+1} = \sum D_t / n$$

$$\text{Or } F_{\text{April}} = \left[\frac{38+27+42}{3} \right]$$
$$= 35.7 \approx 36 \text{ aprons.}$$

Problem:

Suppose the demand for a product in different months (January-March) is given. How many products should they expect to sell in April?

Month	Demand
January	50
February	60
March	70

Time Series Forecasting Models

(3) Exponential Smoothing:

- It uses a special weighted average procedure to obtain a forecast. It provides good forecast results.
- To make a forecast, 3 items are needed:
 1. Current period's forecast
 2. Current period's actual value
 3. Value of a smoothing coefficient, α , which varies between 0 and 1.

$$F_{t+1} = \alpha D_t + (1 - \alpha)F_t$$

Where, F_{t+1} = Forecast of demand for next period, t+1

D_t = Actual value for current period, t

F_t = Forecast for current period t

α = Smoothing coefficient

Time Series Forecasting Models

Question:

Shakti uses exponential smoothing to forecast monthly usage of coffee cream. Its forecast for May was 24 gallons, but the actual usage turned out to be 28 gallons of coffee cream. The manager has chosen to use an alpha of 0.70. what is her forecast for June?

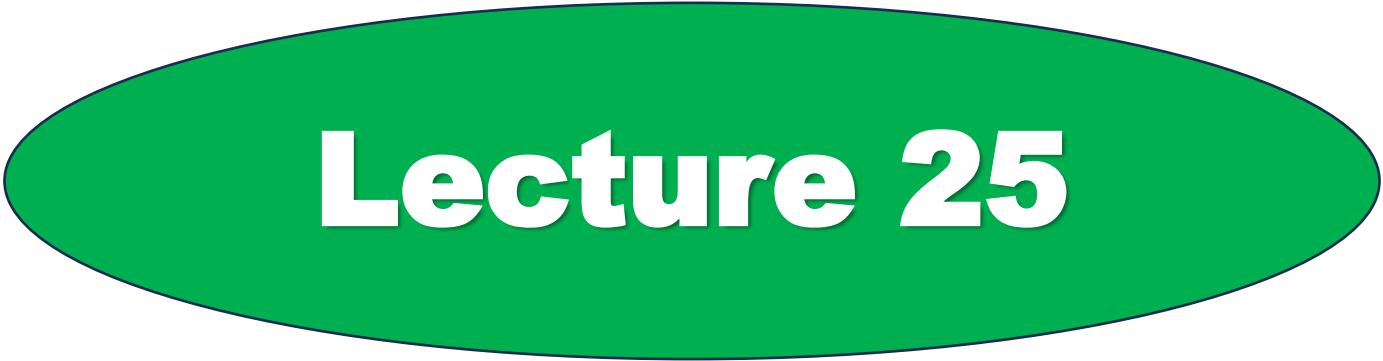
Answer:

Using the exponential smoothing equation

$$\begin{aligned} F_{\text{June}} &= \alpha D_{\text{May}} + (1 - \alpha) F_{\text{May}} \\ &= (0.70)(28) + (0.30)24 \\ &= 26.8 \text{ gallons} \end{aligned}$$

Problem:

The demand for a product in January was 500 units. The forecast for January was 480 units. The smoothing constant is 0.4. Using exponential smoothing, calculate the forecast for February.



Lecture 25

Causal Models

Linear Regression:

- It assumes a linear, or straight line, relationship between two variables.
- The variable being forecast, called the dependent variable, is linearly related to another variable, called the independent variable.
- When we fit the data, the line must be a straight line.

$$Y = a + bX$$

Where, Y= Dependent variable,
X= Independent variable,
a = Y intercept of the straight line,
b= slope of the straight line

Compute Parameter

- Using the linear regression requires first developing the linear regression equation.
- Any value of the independent variable, we can compute the value of the dependent variable, which is the variable being forecast.

Step-1: Compute parameter b

$$b = \frac{\sum XY - n\bar{X}\bar{Y}}{\sum X^2 - n\bar{X}^2} \quad \text{or} \quad b = \frac{n \sum XY - (\sum X)(\sum Y)}{n \sum X^2 - (\sum X)^2}$$

Where, $\bar{Y}$ = average of the Y values

$\bar{X}$ = average of the X values

n = number of data points

Compute Parameter

Step 2: Compute parameter a

$$a = \bar{Y} - b\bar{X} \quad \text{or} \quad a = \frac{\sum Y - b \sum X}{n}$$

Step 3: Substitute values for a and b in the equation

$$Y = a + bX$$

Step 4: Generate a forecast for the dependent variable (Y)

Substitute the appropriate value for the independent variable(X).

Example of Forecasting with Linear Regression

Question:

Pizza boy is trying to forecast the volume of pizza sold based on the amount of money it has put into advertising in the local newspaper. It has been tracking the relationship between pizza sales and advertising over the past four months. The results are as follow:

Pizza Sales	Advertising Dollars
58	135
43	90
62	145
68	145

What would happen to pizza sales if Pizza boy invested \$150 in advertising for next month?

Example of Forecasting with Linear Regression

Answer:

In this example pizza sales are the dependent variable (Y) and advertising dollars the independent variable (X).

	Y	X	XY	X ²	Y ²
	58	135	7830	18225	3364
	43	90	3870	8100	1849
	62	145	8990	21025	3844
	68	145	9860	21025	4624
Total	231	515	30550	68375	13681

$$\bar{X} = \frac{515}{4} = 128.75 \text{ and } \bar{Y} = \frac{231}{4} = 57.75$$

Step 1 Compute parameter b:

$$b = \frac{\sum XY - n\bar{X}\bar{Y}}{\sum X^2 - n\bar{X}^2} = \frac{30550 - 4(128.75)(57.75)}{68375 - 4(128.75)^2} = 0.391$$

Step 2 Compute parameter a:

$$a = \bar{Y} - b\bar{X} = 57.75 - (0.391)128.75 = 7.41$$

Step 3 Substitute values for a and b in the equation:

$$Y = a + bX = 7.41 + 0.391X$$

Step 4 Generate for the dependent variable (Y):

Substitute the appropriate value for the independent variable (X).

$$Y = 7.41 + 0.391(150) = 66 \text{ pizzas}$$

Example of Forecasting with Linear Regression

Problem:

A coffee shop wants to forecast monthly sales based on advertising expenditure. What will be the expected sales if the shop spends Rs. 60 on advertising.

Advertising (Rs)	Sales (Thousand)
20	40
30	50
40	65
50	70

Measuring Forecast Accuracy

- Measuring forecast accuracy is a critical aspect of forecasting.
- Data can change over time, and models that once provided good results may no longer be adequate.
- The model's accuracy can be assessed only if forecast performance is measured over time.
- The first step in measuring forecast accuracy is to measure the forecast error.
- Forecast error is the difference between actual demand and the forecast for a given period:

$$e_t = D_t - F_t$$

Where, e_t = Forecast error for period t

D_t = Actual demand for period t

F_t = Forecast for period t

Measuring Forecast Accuracy

- However, computing error for one time period does not provide enough information.
- Two of the most commonly used error measures are the Mean Absolute Deviation (MAD) and the Mean Square Error (MSE).
- MAD is the average of the sum of the absolute error:

$$\text{MAD} = \frac{\sum |Actual - Forecast|}{n}$$

- MSE is the average of the squared error:

$$\text{MSE} = \frac{\sum (Actual - Forecast)^2}{n}$$

Measuring Forecast Accuracy

Question:

Harry Thurber Apparel is comparing the accuracy of two methods that it has used to forecast monthly sales of its popular jeans. Forecasts of monthly sales from January through May for method A and method B are shown below against the actual sales. Which method provided better forecast accuracy?

Month	Sales	Method A				Method B			
		Forecast	Error	Error	Error ²	Forecast	Error	Error	Error ²
Jan	40	42	-2	2	4	44	-4	4	16
Feb	28	29	1	1	1	31	-3	3	9
Mar	41	39	2	2	4	38	3	3	9
Apr	41	38	3	3	9	42	-1	1	1
May	39	41	-2	2	4	40	-1	1	1
Total			2	10	22		-6		36

Accuracy for method A

$$\text{MAD} = \frac{\sum |\text{Actual} - \text{forecast}|}{n} = \frac{10}{5} = 2$$

$$\text{MSE} = \frac{\sum (\text{Actual} - \text{Forecast})^2}{n} = \frac{22}{5} = 4.4$$

Accuracy for method B

$$\text{MAD} = \frac{\sum |\text{Actual} - \text{forecast}|}{n} = \frac{12}{5} = 2.4$$

$$\text{MSE} = \frac{\sum (\text{Actual} - \text{Forecast})^2}{n} = \frac{36}{5} = 7.2$$

Of the two methods, method A produced both a lower MAD and a lower MSE, which means that it provided better forecast accuracy.

Measuring Forecast Accuracy

Problem:

- A company forecasted sales for 5 months. The actual and forecasted sales (in units) are given below; then calculate

1. Mean Absolute Deviation (MAD)

2. Mean Squared Error (MSE)

Month	Actual Sales (A)	Forecast Sales (F)
1	100	110
2	120	115
3	130	140
4	150	145
5	160	155

Thank You...